Graphene-Analogues Boron Nitride Nanosheets Confining Ionic Liquids: A High-Performance Quasi-Liquid Solid Electrolyte

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Solid electrolytes are one of the most promising electrolyte systems for safe lithium batteries, but the low ionic conductivity of these electrolytes seriously hinders the development of efficient lithium batteries. Here, a novel class of graphene-analogues boron nitride (g-BN) nanosheets confining an ultrahigh concentration of ionic liquids (ILs) in an interlayer and out-of-layer chamber to give rise to a quasi-liquid solid electrolyte (QLSE) is reported. The electron-insulated g-BN nanosheet host with a large specific surface area can confine ILs as much as 10 times of the host's weight to afford high ionic conductivity (3.85×10^{-3} S cm^{-1} at 25 °C, even 2.32×10^{-4} S cm^{-1} at –20 °C), which is close to that of the corresponding bulk IL electrolytes. The high ionic conductivity of QLSE is attributed to the enormous absorption for ILs and the confining effect of g-BN to form the ordered lithium ion transport channels in an interlayer and out-of-layer of g-BN. Furthermore, the electrolyte displays outstanding electrochemical properties and battery performance. In principle, this work enables a wider tunability, further opening up a new field for the fabrication of the next-generation QLSE based on layered nanomaterials in energy conversion devices.

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1. Introduction

Solid electrolytes are the most attractive alternative to liquid carbonate-based electrolytes to improve the safety of rechargeable lithium batteries because they prevent leakage, volatilization, and flammability.[1] Solid electrolytes generally include inorganic and polymer solid electrolytes.[2] Although much endeavor has been devoted to improve those electrolytes, they still suffer from relatively poor ionic conductivity at low temperatures.[3] Another challenge in terms of the application of the solid electrolytes in lithium batteries comes from high interface resistance related to an instable contact between the solid electrolyte and the electrode. In past decade, much endeavor has been made to address these issues but it is still difficult to acquire both high ion conductivity and a good contact between electrode/electrolyte interfaces, when only solid electrolyte materials are used in battery systems.[4]

Ionic liquids (ILs) have been attracted much interest for their unique characteristics such as negligible vapor pressure, nonflammability, good thermal stability and wide electrochemical windows.[5] Incorporation of ILs into polymers or inorganic materials can effectively improve the ionic conductivity of the solid electrolytes, as well as distinctly reduce the interface resistance between the electrode and the electrolyte. Typically, polymer electrolytes consisting of lithium salt, ILs and various polymer hosts such as poly(ethylene oxide),[6] PVdF-HFP,[7] and polymerized ILs (PILs)[8] have been synthesized and applied in batteries at elevated temperatures. Although ILs in these polymer electrolytes increase the ionic conductivity and decrease the electrode/electrolyte interface resistance, the ionic conductivity of them at room temperature is still low and the battery performance is not favorable due to discontinuous transport channels of lithium ions, which is limited by a little uptaking concentration of ILs. Recently, inorganic hybrid solid electrolytes were developed, which consisted of lithium salt, IL and porous SiO_2 or TiO_2 nanoparticles.[9] The nanoparticles can provide an effective pathway for the transport of lithium ions by their self-assembly, suggesting a facile strategy to process inorganic materials into solid electrolytes. To date, the development of versatile matrixes to incorporate ILs for next generation solid electrolytes is still highly desired.

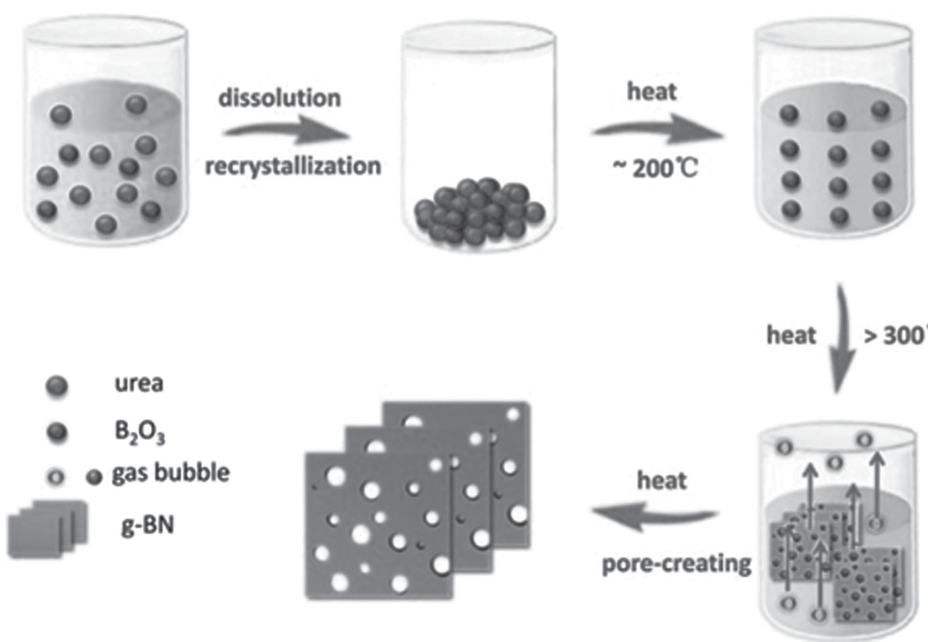
Recently, researchers have been harnessing the properties of 2D graphene-analogue boron nitride (g-BN) nanosheets,[10] such as a wide energy band gap,[11] high thermal conductivity,[12] good chemical stability,[13] high surface area,[14,15] and excellent adsorption performance.[16] Toward the study of safe electrolytes, g-BN intrinsically possesses several unique properties: i) the electron insulating feature of g-BN makes it an appropriate solid electrolyte host without the possibility of short circuit; ii) the rich porosity and robust mechanical modulus enables g-BN to incorporate a large amount of ILs, yet still keeping a solid state[17]; iii) the uniform and wide channels between or outside g-BN layers can lead to a smooth transport pathway for lithium ions, thus increasing the ionic conductivity even at a solid state. Inspired by the above design rationale, herein, the few-layered g-BN with a highly abundant nanoporous structure has been developed as an electrolyte host. A quasi-liquid solid electrolyte (QLSE) is defined as solid state with liquid electrolyte merits. Such materials confining with a large quantity of ILs not only represent solid electrolytes state, but also possess both advantages of solid and liquid electrolytes. The QLSE displays the exceptional ionic conductivity (more than 10^{-3} S cm^{-1}) at room temperature, the low interface resistance and improved interface compatibility. Furthermore, batteries derived from the above QLSE display a high capacity and good cycling stability.

2. Results and Discussion

2.1. Fabrication of g-BN Nanosheets and QLSE

We explored a synthesis route of nanoporous BN nanosheets depending on a template-free method, as illustrated in **Scheme 1**. The mixture of boron oxide (B_2O_3) and urea (CON_2H_4) in ethanol aqueous was recrystallized. Then the obtained mixture was heated gradually in tube furnace under nitrogen. Based on the intricate urea pyrolysis procedure,[18] we propose the formation process of few-layered nanoporous g-BN. When the temperature was raised to 200 °C, urea transferred to liquid and polycondensated to the intermediate cyanuric acid ($C_3H_3N_3O_3$).[18] Urea melted into the liquid phase, functionalizing not only as solvothermal solvent to promote the molecular diffusion but as the peeling role to control formation of few-layered g-BN.[19,20] Increasing the temperature continuously, g-BN grows up. During this process, urea and formed urea intermediates decomposed to NH_3, CO_2, and H_2O bubbles which play a pores-creating and exfoliating role for the resultant nanoporous few-layered structure. Finally, few-layered and nanoporous g-BN could be fabricated.

The g-BN nanosheets with nanoporous structure have been verified by a BET measurement. **Figure 1**a,b indicates nitrogen



Scheme 1. Schematic of the synthesis process of g-BN.

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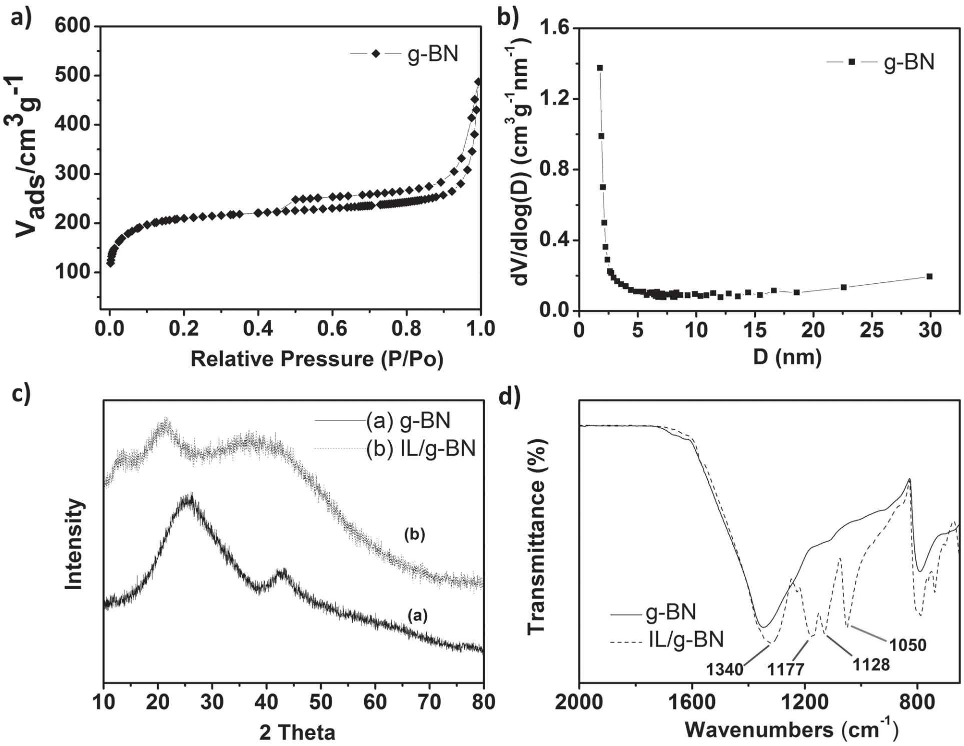


Figure 1. Structure and surface property of g-BN and QLSE. a) N_2-desorption and sorption isotherms of g-BN. b) BJH pore size distribution. c) XRD patterns of g-BN and QLSE. d) FT-IR spectra of g-BN and QLSE.

isotherms and pore size distribution curves of the g-BN, which can be categorized as type II based on the IUPAC classification, and type H3 hysteresis loop in the relative pressure range of 0.45–1.0 indicates the g-BN corresponds to nanosheets materials.[21] Additionally, very Low pressure adsorption presents a micropore-filling behavior. Moreover, the type H3 hysteresis loop at a relative pressure of 0.45 to 1.0 suggests the presence of slits-shaped pores, in line with the shape of the sheet-like nanostructures. The total BET surface area, micropore surface area, micropore volume, and pore diameter of the g-BN are 860 $m^2 g^{-1}$, 602 $m^2 g^{-1}$, 0.2 $cm^3 g^{-1}$, and 2 nm, respectively. High surface area and the surface micropore nanostructure contribute to confine ultrahigh concentration of IL (tenfold Li-IL) in g-BN to form a QLSE.

The X-ray diffraction (XRD) patterns of both g-BN and QLSE are shown in Figure 1c. The two peaks corresponding to the (002) and (100) planes of g-BN are readily in accordance with the standard hexagonal phase of *h*-BN (JCPDS Card No. 34–0421), demonstrating that the as-obtained g-BN has a hexagonal structure.[12] When the g-BN adsorbs Li-IL, an obvious shift from $2\theta = 24°$ to low angles can be observed, indicating that lithium ions may be embedded in the interlayer space of the g-BN.[21] The g-BN confining Li-IL was further determined by FT-IR spectrum in Figure 1d. In the case of g-BN, the absorption band centered around 1360 cm^{-1} and 795 cm^{-1} can be attributed to the in-plane B-N transverse optional mode and out-of-plane B-N-B bending mode of hexagonal boron nitride.[22] Compared with QLSE, four obvious peaks in the appeared (1340, 1177, 1128, and 1050 cm^{-1}) that could be assigned to the vibration sorption of TFSI anions of ILs ($\nu_a(SO_2)$, $\nu_a(CF_3)$, $\nu_s(SO_2)$, and ν_a(S-N-S),[23] respectively, which demonstrated that g-BN confining ultrahigh concentration of Li-IL.

In order to further investigate the formation of continuous transport channels of lithium ions, the microscopic performance of the g-BN and QLSE samples were used to determine. From the Scanning electron microscopy (SEM) images (Supporting Information Figure S1), the nanosheets of the g-BN can be clearly observed, and after absorbing the Li-IL solution they still exist around the groups of nanosheets of the g-BN. The lower magnification STEM images in **Figure 2**a,d also verify that g-BN and QLSE maintained a graphene-like flake morphology.[13] Higher magnification images in Figure 2b,e further reveal the polycrystalline feature of the g-BN with few-layers structure (3–4 layers). Moreover, the hexagonal structure of g-BN (Figure 2c,f) can also be seen clearly from the lattice fringes. After confining the IL solution, QLSE still remains graphene-like layered structure (Figure 2e) and the width of polycrystalline fringe increases compared to the g-BN (Figure 2e), indicating the distance of nanolayers is wider due to the insertion of lithium ions, which is in accordance with the XRD results. Furthermore, it is also seen in Figure 2f, the indistinct STEM image indicates that the IL solution was confined in nanopores and dispersed on the surface of the layered g-BN. Thus the transport channel for lithium ions in interlayer and out-of-layer can be developed. **Scheme 2** illustrates the fabrication of the QLSE and the pathway for lithium ions transport formed by the Li-IL solution in the interlayer and out-of-layer of g-BN nanosheets. The continuous transport channels are favorable for the fast-transport of lithium ions and may result in the high ionic conductivity.

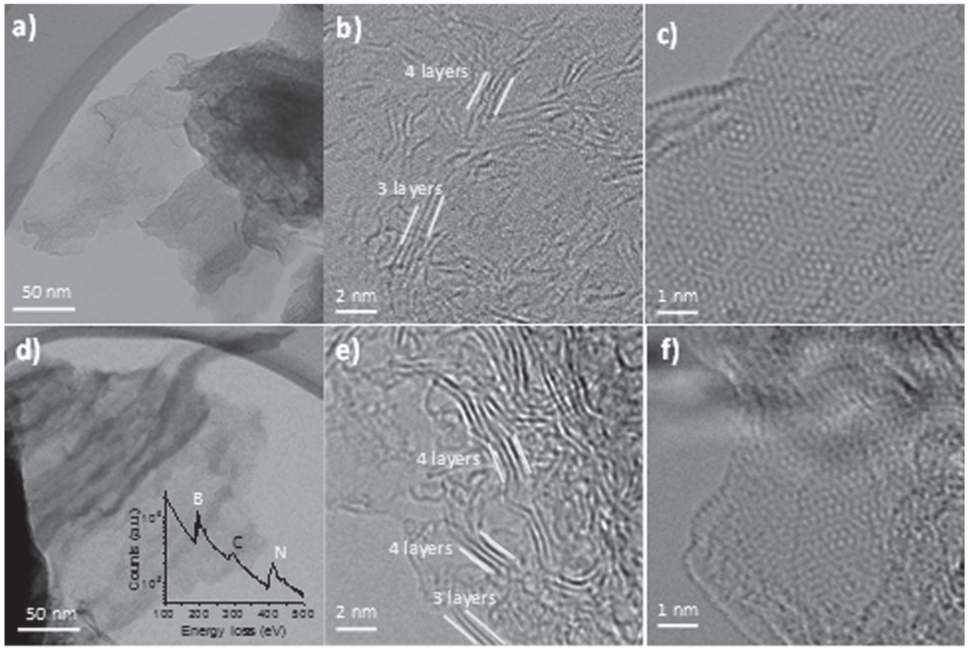
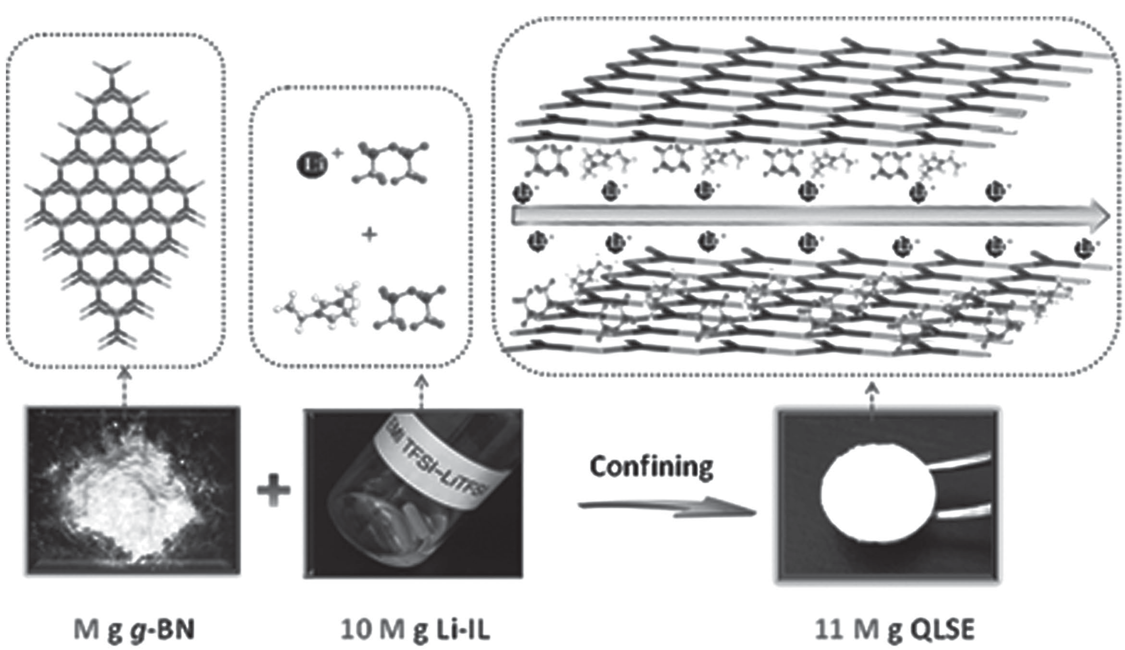


Figure 2. Representative STEM bright field (BF) images. a–c) g-BN and d-f) QLSE samples (mass ratio of IL and g-BN is 2). Lower magnification images in panels (a) and (d) show that BN maintained a flake morphology after the absorption of Li-IL. Higher magnification images in panels (b) and (e) reveal the layer structure at the edges of the flakes, where the materials appeal to be curved into edge-on condition (white arrows). The EELS spectrum of the QLSE specimen was overlaid with (d) showing the B and N K-edges and C in IL. These images also reveal the poly-crystalline nature of the material. The hexagonal BN structures for both materials can be seen from the lattice fringes shown in panels (c) and (f).

2.2. Measurement of Electrochemical Properties

To verify the above hypothesis, the ionic conductivity at different temperatures has been characterized. **Figure 3**a shows the result of ionic conductivity for the QLSE and the Li-IL electrolyte control. The plot of QLSE as the same with the Li-IL electrolyte exhibits the Vogel–Tamman–Fulcher-type temperature dependence, which indicates the value of ionic conductivity of QLSE correlates with the viscosity of the solution and the hopping species in QLSE are determined by the property of Li-IL electrolyte.[24] It should be noted the ionic conductivity of QLSE is very high at room temperature that reaches 3.85×10^{-3} S cm^{-1}, even though at –20 °C it still is 2.32×10^{-4} S cm^{-1}. The high ionic conductivity is attributed to the enormous adsorption for ILs and the ordered lithium ion transport channels in g-BN nanolayers which are formed when the Li-IL fills in, as observed by STEM. The outstanding property of QLSE at lower temperatures opens a



Scheme 2. QLSE pathway for lithium ion transport formed by the IL solution in the interlayer and out-of-layer of g-BN.

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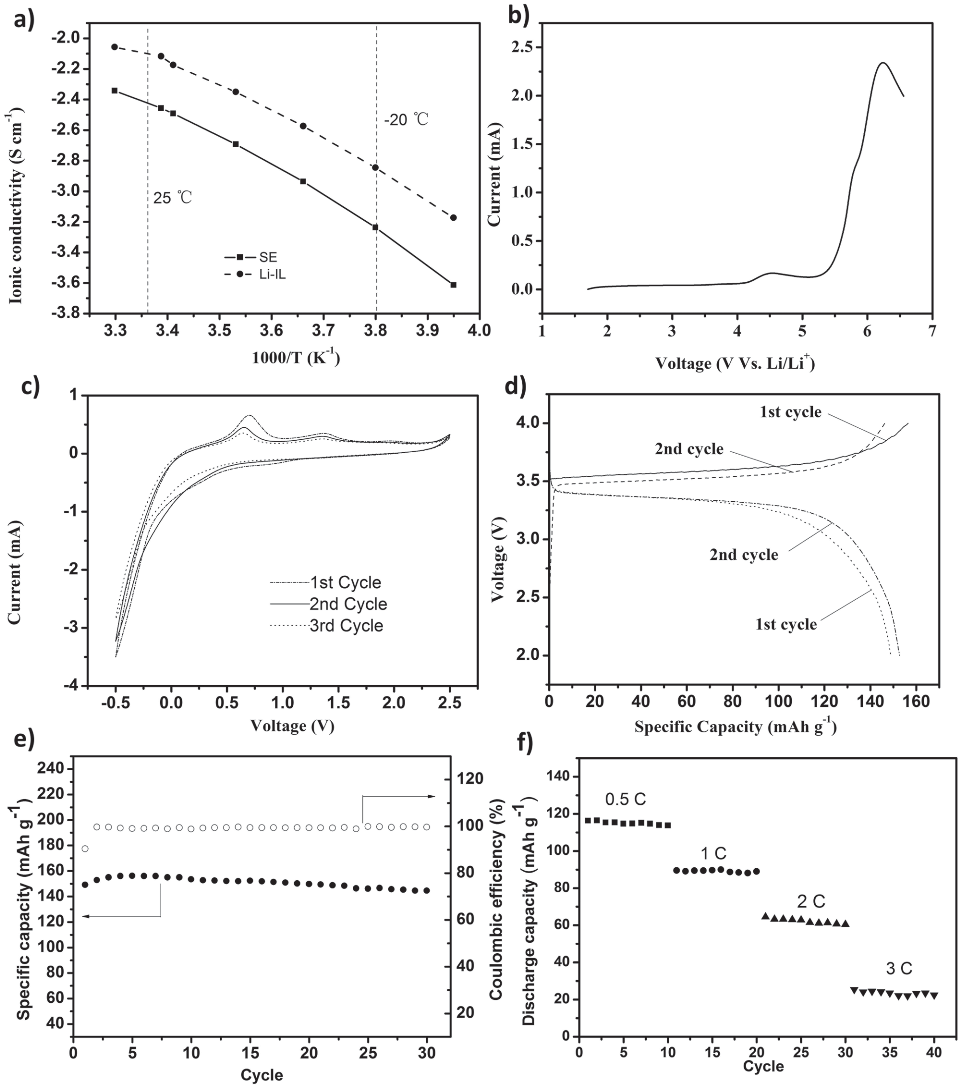

Figure 3. Electrochemical performance of QLSE. a) The temperature dependence of ionic conductivity for the QLSE and Li-IL electrolytes. b) The linear sweep voltammetry curve with a Li/QLSE/SS cell at room temperature; SS: stainless steel. c) Cyclic voltammograms with a Li/QLSE/Pt cell. Working electrode: Pt; counter electrode and reference electrode: lithium; scan rate: 10 mV s^{-1}. d) Voltage versus discharge capacity profile of a Li/QLSE/$LiFePO_4$ cell. e) Discharge capacity as a function of cycle number for a Li/QLSE/$LiFePO_4$ cell. Charge–discharge current rate is 0.1 C. f) Rate capacity against the cycle number for a Li/QLSE/$LiFePO_4$ cell at different charge/discharge rates. Charge current rate is fixed at 0.1 C.

promising perspective as a new QLSE in application to low temperatures for lithium batteries.

The electrochemical stability is an important index which determines the availability of electrolytes. The electrochemical stability of the QLSE was characterized by linear sweep voltammetry (LSV) with a Li/QLSE/SS cell at room temperature, as shown in Figure 3b. A peak appears at about 4.5 V versus Li/Li which is associated with the oxidation decomposition of imidazolium cations in the IL. The oxidation stability of the QLSE agrees with that of the IL, which is 4.3 V (corresponding to a current density of 10 mA cm^{-2}). This result demonstrates that the electrolyte has good electrochemical stability, thus confirming its feasibility for application in Li/$LiFePO_4$ batteries.

Lithium redox in the QLSE was characterized by cyclic voltammograms (CVs) with a Li/QLSE/Pt cell, as shown in Figure 3c. The plating of Li on the nickel electrode could be clearly observed like EMI-TFSI IL electrolytes.[17] In the first cycle for the QLSE, the cathodic peak corresponding to the plating of Li is about –0.12 V versus Li/Li^+, and in the returning scan the anodic peak corresponding to the stripping of Li is around 0.15 V versus Li/Li^+. It is noteworthy that in case of the Pt electrode, the peak at 0.7 V versus Li/Li^+ due to the formation of an Li-Pt alloy could be clearly observed.[25,26] The Li redox in the QLSE could be generated by forming a certain surface film (SEI) on the Pt electrode. The redox peak currents decrease gradually with the cycle numbers. This suggests that the SEI film turns thicker and thicker so that the lithium reduction and oxidation is restrained gradually.

The charge–discharge performance of Li/QLSE/$LiFePO_4$ batteries was characterized at room temperature, and their cycling properties are presented in Figure 3e. The discharge capacity and the coulombic efficiency of the battery are found to be 149 mAh g^{-1} and 90.5% in the first cycle, respectively. During the initial 4 cycles, the discharge capacity and the coulombic efficiency increased gradually, perhaps as a result of generation of improved penetration and contact of the IL component from the electrolyte into the electrode material. After an increasing step, the cell delivers a maximum capacity of about 155 mAh g^{-1}, and the coulombic efficiency reaches 99%. Subsequently, the battery capacity decreases slowly during cycling. Additionally, the rate performance of batteries has been determined. From the Figure 3f, we can know batteries display acceptable rate capacities. When operated at 0.5, 1, 2, and 3 C, the discharge capacities retain 115, 89, 62, and 24 mAh g^{-1}, respectively. The decrease of the capacity with increasing the rate mainly arises from high viscosity and multiple charged ions of the Li-IL. It is worthy noted that the battery performance is greatly associated with the thickness of the QLSE membrane and the shaping stress of the membrane. The optimizing technology of the fabrication for QLSE membranes is ongoing.

2.3. Nanoconfining Effect for Lithium Ion Transport

The g-BN nanosheets have high specific surface area which is attributed to its nanoporous and few-layered structure. When the small amount of Li-IL is supported on g-BN nanosheets, the Li-IL could cover the outside surface of the nanolayers. As the amount of the Li-IL increases, the Li-IL gradually enters the inter-layer and is confined in nanopores space of g-BN, which is demonstrated by XRD patterns for different contents of the Li-IL showed in **Figure 4**. The (002) diffraction peak hardly shifts when the content of the Li-IL is less than 0.4-fold of the weight ratio of g-BN, and afterward a clear shift of the (002) diffraction peak to low angles is observed and the intensity of the peaks increases obviously as the content of the Li-IL increases, which verified the distance between the (002) basal planes is broadened. Thus, it could be inferred the pathway for Lithium ions transport in QLSE consists of two parts, the outside surface space and the inter-layer space of g-BN, due to the confining numerous Li-IL.

Additionally, the content of ILs greatly affects the electrochemical performance of the QLSE. From the Supporting Information Figure S3, the temperature dependence of ionic conductivity for the different contents of the QLSE, we can know the ionic conductivity increases apparently as the content of ILs, and it more and more approaches the value of the Li-IL electrolyte. Another advantage of is that the interface resistance between the QLSE and electrodes decreases and the interface compatibility improves, as the content of Li-IL increases.[27]

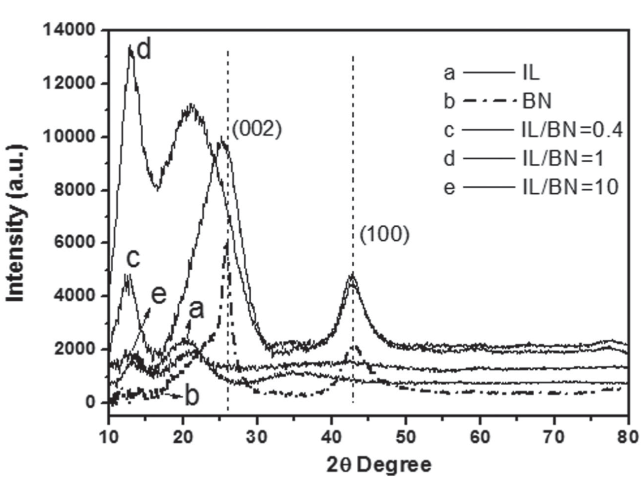

Figure 4. XRD patterns for different contents of the Li-IL in g-BN nanosheets.

3. Conclusions

In conclusion, nanoporous graphene-analogues g-BN nanosheets have been synthesized using a thermal treatment process without template. The obtained g-BN nanosheets have a very high specific surface area, and they could absorb the Li-IL solution as much as tenfold of its own weight. The QLSE has excellent ionic conductivity at room and lower temperature, and it is 3.85×10^{-3} S cm^{-1} at 25 °C and 2.32×10^{-4} S cm^{-1} at –20 °C. The distinguished property of QLSE at lower temperatures opens a promising perspective as a new QLSE in application to low temperatures for lithium batteries. The high ionic conductivity is resulted from the enormous absorption for ILs and the ordered Lithium ions transport pathways between and in g-BN nanolayers, which are developed as the Li-IL solution confining effect. This work explores a new field for the application of layered nanomaterials as electrolyte host in energy conversion devices.

4. Experimental Section

Synthesis: Preparation of nanoporous g-BN nanosheets: 1.64 g boron oxide (B_2O_3) and 24 g urea(CON_2H_4) were dissolved in a mixture of 40 mL ethanol and 20 mL ultrapure water. The solution was stirred and heated at 70 °C to remove solvents for recrystallization. Then the white solid was transferred to a tube furnace, and heated to 900 °C at a heating rate of 4 °C min^{-1} under N_2 atmosphere. After the temperature reached 900 °C, the tube furnace was kept for another 2 h. Then, the tube furnace freely cooled to room temperature, and finally the g-BN nanosheets were obtained.[11,15]

The IL, EMI-TFSI, was prepared according to a reference method.[26] A lithium salt-IL solution of molality 0.5 mol kg^{-1} (Li-IL) was prepared by adding a proper amount of LiTFSI to the IL. The water content of EMI-TFSI was measured by coulometric Karl Fischer titration to be 20 ppm.

The QLSE was prepared by g-BN nanosheets and the Li-IL mixed simply with the weight ratio of 1:10 in argon atmosphere and the thickness is 0.35 mm, noted as QLSE. The ratio in the electrolyte has been optimized so as to obtain Li-IL as much as possible. This sample was used during all the electrochemical tests except battery performance.

Characterization: XRD patterns of the samples were collected using a Panalytical X'Pert PRO diffraction system with a Cu Kα radiation. SEM was performed by a Zeiss Supra 55 VP SEM. Nitrogen adsorption and desorption isotherms were obtained using a Tristar 3000 apparatus at 77K. FTIR spectra were obtained with a Nicolet 7199 FTIR spectrometer. The specimen for electron

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microscopy was prepared by grinding and dispersing dry powders of the material onto 400 mesh holey carbon copper grid. The Scanning transmission electron microscopy (STEM) image was performed on Nion UltraSTEM 100 (operated at 100 kV) in Oak Ridge National Laboratory.

Electrochemical Measurements: The ionic conductivity of the QLSE was determined from the impedance spectrum using a blocking cell of which the electrolyte was sandwiched between two stainless steel electrodes in a Swagelok cell. Electrochemical impedance spectrum measurements were performed using a VMP3 (BioLogic) over a frequency range of 1 MHz to 100 mHz with a potentiostatic signal amplitude of 10 mV. LSV was performed at 25 °C (scan rate 10 mV s^{-1}) using a Pt/QLSE/Li Swagelok cell.

Battery Test: Lithium foil (battery grade) was used as a negative electrode. And the positive electrode was fabricated by spreading the mixture of $LiFePO_4$, acetylene black, and PVdF (initially dissolved in *N*-methyl-2-pyrrolidone) with a weight ratio of 8:1:1 onto Al current collector (battery use). Loading of active material was about 2.0 mg cm^{-2} corresponding to 0.3 mAh cm^{-2} and this thinner electrode was directly used without pressing.

The QLSE used in batteries was fabricated by a squash technique. g-BN and Li-IL with a weight ratio of 1:5 were dispersed in ethanol for 8 h. Then the solvent was removed at 80 °C, and the obtained white powder was pressed into pellets with the diameter of 12 mm and the thicknesses of 0.35 mm. The as-prepared pellets were dried at 110 °C under vacuum for 24 h. Li/$LiFePO_4$ polymer batteries were fabricated (in an argon-filled glove box) by laminating the lithium foil, the QLSE and an $LiFePO_4$ cathode tape in a button cell. Preliminary cycling tests on Li/$LiFePO_4$ polymer batteries were performed at 25 °C using a CT2001A cell test instrument (LAND Electronic Co., Ltd.). The charge and discharge current rates were fixed to C/10. The voltage cut-offs were fixed at 4.0 (charge step) and 2.0 V (discharge step), respectively.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

M.T.L. and W.S.Z. contributed equally to this work. S.D., M.T.L., and W.S.Z. conceived the idea. W.S.Z. and Y.H.C. carried out the BN synthesis and characterization. M.T.L. prepared the quasi-liquid solid electrolytes and performed the electrochemical test. P.F.Z., B.L.Y., and H.M.L. analyzed the physicochemical properties of this new electrolyte. Q.H. and A.B. performed the STEM measurement and data analysis. M.T.L. and W.S.Z. wrote the paper. M.T.Li, W.S.Z., P.F.Z., and S.D. discussed the results and participated in the preparation of the paper. M.T.Li, W.S.Z., and Y.H.C. appreciate the financial support from the National Natural Science Foundation of China (Nos. 21376111, 21303132, 21576122 and 21506083) and Six Big Talent Peak in Jiangsu province (JNHB-004). P.F.Z. and S.D. were supported by the US Department of Energy, Office of Science, Office of Basic Energy Sciences, Division of Materials Science and Engineering. Electron microscopy research (QH and AYB) is supported by the U.S. Department of Energy, Office of Science, Basic Energy Sciences, Materials Sciences and Engineering Division and through a user project supported by ORNL's Center for Nanophase Materials Sciences, sponsored by the Scientific User Facilities Division, Office of Science, Basic Energy Sciences, U.S. Department of Energy.

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